SALARY PREDICTOR

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Introduction

In today's competitive job market, understanding salary expectations is a major challenge for students and job seekers. Many individuals are unsure about how factors such as experience and education influence their earning potential. This often leads to poor career decisions, underpayment, or unrealistic expectations.

This project aims to address this issue by building a simple Machine Learning model that predicts salary based on key factors. It demonstrates how AI/ML can be applied to solve real-world problems in a practical and understandable way.

Problem Statement

The main problem addressed in this project is:

"To predict the expected salary of an individual based on their years of experience and education level."

The goal is to provide a data-driven estimation that can help users make better career and financial decisions.

The system should:

- **Accept user input (experience and education)**
- **Process the input data**
- **Predict salary using a trained model**
- **Display the result clearly**

Functional Requirements

The system must perform the following functions:

- **Load dataset from a CSV file**
- **Preprocess data (handle categorical values)**
- **Train a Machine Learning model**
- **Accept user input from the console**
- **Predict salary based on input**
- **Display predicted salary output**
- **Show graphical representation (salary vs experience)**
- **Calculate and display model accuracy**

Non-Functional Requirements

The system should satisfy the following:

- **Usability: Easy to run and understand**
- **Performance: Fast execution with minimal delay**
- **Reliability: Accurate predictions based on dataset**
- **Scalability: Can be extended with more features**
- **Portability: Can run on any system with Python installed**
- **Maintainability: Code should be simple and readable**

Design Diagram

System Flow:

User Input (Experience, Education)

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Data Preprocessing (Encoding)

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Trained ML Model (Linear Regression)

↓

Salary Prediction

↓

Output Display + Graph

Implementation Details

The project is implemented using Python and follows these steps:

Dataset Loading

- **Dataset is stored in a CSV file (salary_data.csv)**
- **Loaded using Pandas**

Data Preprocessing

- **Education column (categorical) is converted into numerical values using Label Encoding**

Model Training

- **A Linear Regression model is used**
- **Input features:**
  - **Experience**
  - **Encoded Education**
- **Output:**
  - **Salary**

Prediction

- **User enters:**
  - **Years of experience**
  - **Education level**
- **Model predicts salary based on trained data**

Visualization

- **Graph plotted using Matplotlib:**
  - **X-axis → Experience**
  - **Y-axis → Salary**

Screenshot / Results

Sample Output 1:

Experience: 3
Education: Bachelor
Predicted Salary: ₹42000

Sample Output 2:

Experience: 6
Education: Master
Predicted Salary: ₹72000

Sample Output 3:

Experience: 9
Education: PhD
Predicted Salary: ₹115000

Graph Output:

- **A scatter plot showing salary vs experience**
- **A trend line representing Linear Regression**

Testing Approach

The project was tested using:

Unit Testing

- **Verified dataset loading**
- **Checked encoding functionality**

Input Testing

- **Tested with different experience values**
- **Tested all education categories**

Output Validation

- **Compared predicted values with dataset trends**

Edge Case Testing

- **Tested with minimum and maximum experience values**

Challenges Faced

- Handling categorical data (education encoding).
- Limited dataset size.
- Ensuring correct input format from user.
- Avoiding overly simplified predictions.
- Understanding model accuracy metrics.

Learnings & Key Takeaways

- Gained understanding of Machine Learning basics
- Learned how Linear Regression works
- Understood data preprocessing techniques
- Learned how to convert categorical data into numerical form
- Gained experience in data visualization

Future Enhancements

Add more features:

- **Skills**
- **Job role**
- **Location**

Use advanced algorithms:

- **Random Forest**
- **Gradient Boosting**
- **Build a web application using Streamlit**
- **Use real-world datasets from platforms like Kaggle**
- **Deploy the project online for public use**

Conclusion

This project successfully demonstrates how Machine Learning can be applied to predict salaries based on simple input features. It provides a practical understanding of regression models and highlights the importance of data-driven decision-making.